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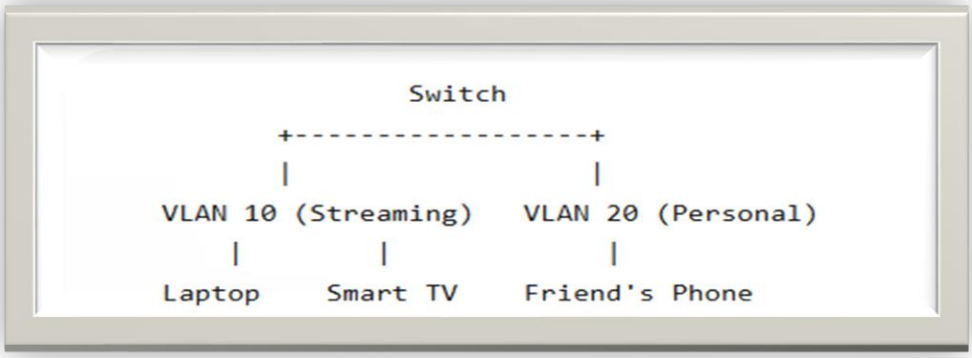
N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

SWITCHING AND VLANS

TASKS:1

Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

RESULT



TASKS:2

Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

RESULT

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CONFIGRATION COMMAND

Switch> enable

Switch# configure terminal

Switch(config)# vlan 10

Switch(config-vlan)# name STREAMING

Switch(config-vlan)# exit

Switch(config)# vlan 20

Switch(config-vlan)# name PERSONAL

Switch(config-vlan)# exit

Switch(config)# interface range fa0/1-3

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 10

Switch(config-if-range)# exit

Switch(config)# interface range fa0/4-5

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config-if-range)# exit

VERIFICATION :

Run: show vlan brief

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TASKS:3

In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

RESULT

BEFORE:

Switch(config)# interface fa0/3

Switch(config-if)# switchport access vlan 10

CHANGE TO VLAN 20:

Switch(config)# interface fa0/3

Switch(config-if)# switchport access vlan 20

VERIFICATION:

show vlan brief

Ping test:

- Before moving → PC3 communicates with VLAN 10.
- After moving → PC3 communicates with VLAN 20.

TASKS:4

Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3). Use a loop or range command if supported by the tool to avoid repeating the same command 10 times

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RESULT

Switch> enable

Switch# configure terminal

Switch(config)# vlan 30

Switch(config-vlan)# name OFFICE

Switch(config-vlan)# exit

Switch(config)# interface range fa0/1-10

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 30

Switch(config-if-range)# end